Week 1: Introduction

Program overview, tooling, and data

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Industry Mentor: Dendi Suhubdy (Bitwyre) · Guest: Fenni Kang (Coincall)

Welcome to the Program

- A 10-week summer research program on optimal market making for crypto options
- Goal: design, implement, and evaluate quoting strategies on real Coincall data
- You will produce a research-grade report and a working code repository
- This week: orientation, environment setup, and data access

Maker vs. Taker

- A market maker continuously quotes both sides and earns the spread
- A taker crosses the spread to execute immediately
- We build the maker's side; in Week 9 a Coincall practitioner covers taker strategies
- The spread is compensation for committing to quotes under uncertainty

The Spine: Avellaneda-Stoikov

- The whole course is organized around the Avellaneda-Stoikov framework
- Introduced Week 3; turned into a calibrated engine Week 5
- Extended to multi-Greek options inventory Week 8
- Advanced extensions (multi-asset, adverse selection, robust) in Week 10

Tooling

- Problem sets: Python + PyTorch (autograd does the differentiation for you)
- You will NOT hand-roll algorithmic differentiation
- Fast-computation week (Week 7): C++ exposed to Python via pybind11
- Industry pattern: Python for research, C++ for the hot path

Data, Milestones, Assessment

- Coincall BTC/ETH order books, trades, options chains, funding rates
- Milestone 1 (Wk5): Avellaneda-Stoikov engine
- Milestone 2 (Wk7): fast pricing & Greeks library
- Milestone 3 (Wk9): full options market-making backtest
- Grade: problem sets 30%, milestones 30%, final 30%, participation 10%

This Week's To-Do

- Set up Python + PyTorch and a C++17 toolchain with CMake
- Build and import the pybind11 'hello world' starter module
- Get your Coincall data credentials and sign the data-use agreement
- No graded problem set this week